Group Project: e-commerce Solution

Due Date: April 10, 2026.

Goal:

Its goal is to facilitate the production of a system that supports the business needs, business requirements, is easy to use, and meets users' needs. Moreover, it must store all the relevant data and provide information in proper formats for customers, employees and vendors.

Purpose: The purpose of this project is to help you

- apply the knowledge you have already gained to solve a concrete problem
- build the communication skills by collaborating with group member(s)
- become familiar with Oracle Database and SQL Server Developer
- become familiar with database access control by granting appropriate privilege to your group member

Read instructions below carefully:

- This project can be completed in maximum group of 4 students
- One group member submits the package through Luminate on behalf of the whole group
- Please make sure that your submission includes ER diagram, script file to create tables and populate sample data, the queries to fulfill business requirements and the output generated by those queries

Rubric

5% - ER model
5% - Table creation
5% - Query (Inserts, queries)

<u>Overview</u>

Database Management System is to support business needs and requirements while ensuring ease of use and meeting user requirements, therefore, it must efficiently store all relevant data and provide information in appropriate formats for different stakeholders, such as business managers, employees, customers, vendors, etc.

Database Design: A Step-by-Step Overview

Database design is a structured process used to create a database that efficiently stores, retrieves, and manages data for an organization.

- 1. Requirement Analysis

  Purpose: Understand the goals, data needs, and scope of the database.

  Key Activities:

  - Identify stakeholders (e.g., end-users, developers, business analysts).
  - Gather functional and non-functional requirements:
    o Types of data to store
    o Expected queries and reports
    o Business rules and constraints
  - Study existing systems (e.g., Amazon.ca, Booking.com)
  - Define key use cases and workflows.

  ❖ Output: Requirements Document

- 2. Conceptual Design

  Purpose: Create a high-level data model that captures essential entities and relationships.

  Key Activities:

  - Develop conceptual Entity-Relationship Diagram (ERD):
    o Entities (e.g., Customer, Order)
    o Attributes (e.g., CustomerName, OrderDate)
    o Relationships
  - Identify primary keys for each entity

  ❖ Output: Conceptual Schema (ERD)

- 3. Logical Design

  Purpose: Translate the conceptual model into a logical model tailored for a relational database.

  Key Activities:

  - Normalize the schema to reduce redundancy and enforce data integrity.
  - Define tables and columns with appropriate data types.
  - Specify constraints:
    o Primary and foreign keys
    o Unique, not null, and check constraints

- ❖ Output: Logical Schema (Normalized Table Structure)

- 4. Implementation

Purpose: Build and deploy the database using Oracle

Key Activities:

- Write and execute SQL scripts to:
  - Create tables, constraints
- Provide sample data, at least 10 rows for each table

- ❖ Output: Operational Database

Business Requirements for the project

Provide queries to accomplish following:

1. Search products based on category
2. Manage shopping cart
3. Register/unregister user
4. Track shipping status
5. Manage vendors
6. Generate monthly report (e.g., sales per month)
7. Identify the most popular product(s)
8. Categorize customers based on his/her purchase history
9. List all items on user's wish list
10. Handle returns and refunds